MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

SERIES 1 • CARD 02 / 12 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS

CARD 02 / 12 • DOMI-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.

It contains all unordered pairs [alb] from 0 to 6:

Total Tiles = T(7) = (7 × 8) / 2 = 28

Unique Tiles

Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ⌀ recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE SUM

[5|4] pips = 9

[3|2] pips = 5

Total: 9 + 5 = 14

T(7) = 28 Unique Tiles

Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC	COLOR
Double-Six Tiles	28x	28×14×7 mm	Stealth Black
Wildcard Blanks	2x	28×14×7 mm	Stealth Black
12-Frame Tray	1x	Dual 6-Bay	Matte Charcoal
Doubles Lid	1x	7-Slot Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Arrange the 28 standard tiles and 2 wildcards into 4 columns of 7 and 1 row of 2 on the floor. Missing tiles break the brick layout! Snap lid securely.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART–WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/math–dominoes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS K.OA.A.3 (Decompose to 10)
- CCSS 1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Additi...)

Grades K–5 Addition & Combinatorics

GUIDING INQUIRY

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

1 STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]=0, [1|1]=2, [2|2]=4, [3|3]=6, [4|4]=8, [5|5]=10, [6|6]=12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

2 STEP 2: PAIR COMPLEMENTARY SUM–TO–6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,

3–4 STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14.

5 CORE PRINCIPLE: THE 28–TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]),
6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ EXTENSION INVESTIGATION: FIND ALL SUM–TO–7 TILES

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.